Name:_____

# Exam Question Booklets
# OCR Physics A

## Booklet 6
## 11 Questions
## 98 marks

## Things to remember:

✓

✓

✓

✓

✓

✓

✓

1 A ski jumper skis down a runway and projects himself into the air, landing on the ground a short time later. The mass of the ski jumper and his equipment is 80 kg. Fig. 5.1 shows the skier just before he leaves the runway where his velocity is 20 m $s^{-1}$ in a horizontal direction.

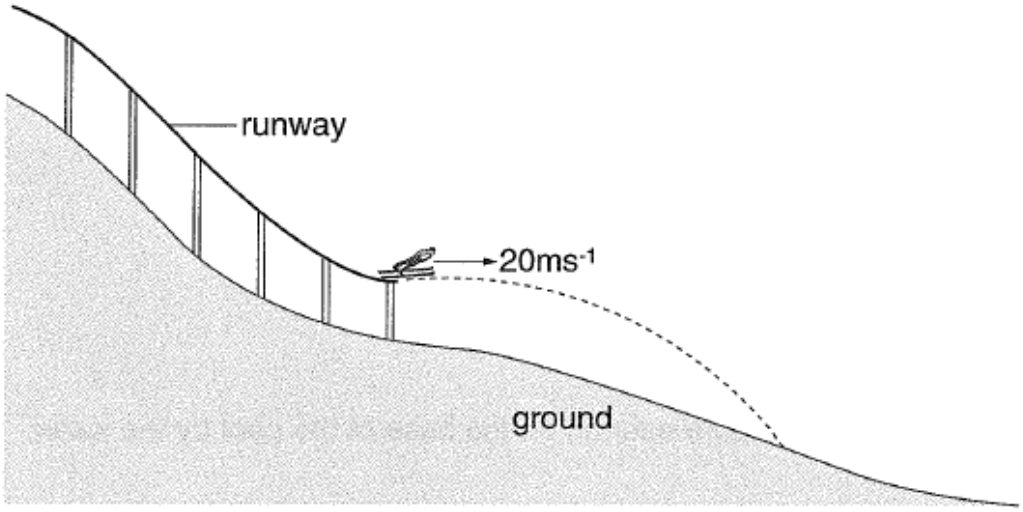


**Fig. 5.1**

**(a)** The skier lands 4.0 s after leaving the runway. Assume that only a gravitational force acts on the skier. Calculate

**(i)** the horizontal distance travelled by the skier in 4.0 s

horizontal distance = ..... m [1]

**(ii)** the vertical fall of the skier in this 4.0 s

vertical fall = ..... m [3]

**(iii)** the horizontal component of the skier's velocity immediately before he lands

horizontal component = .....m $s^{-1}$ [1]

**(iv)** the vertical component of the skier's velocity immediately before he lands.

vertical component = ..... m $s^{-1}$ [2]

**(b)** (In this question, marks are available for the quality of written communication.)

Take **all** the forces that act on the skier into account.

**(i)** State the forces that act on the skier when he is in the air.

.....

.....

..... [2]

(ii) Discuss the energy changes that occur as the skier travels down the runway and explain how the skier attempts to increase the horizontal distance he travels before landing on the ground.

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[7]

2

**(a)** Explain the quantities

**(i)** gravitational potential energy

.....

.....

..... [2]

**(ii)** kinetic energy

.....

.....

..... [2]

**(iii)** power.

.....

.....

..... [1]

**(b)** Water leaves a reservoir and falls through a vertical height of 130 m and causes a water wheel to rotate. The rotating wheel is then used to produce 110 kW of electrical power.

**(i)** Calculate the velocity of the water as it reaches the wheel, assuming that all the gravitational potential energy is converted to kinetic energy.

velocity = ..... $m\,s^{-1}$ [3]

**(ii)** Calculate the mass of water flowing through the wheel per second, assuming that the production of electrical energy is 100% efficient.

mass of water per second = ..... unit ..... [3]

**(iii)** State and explain **two** reasons why the mass of water flowing per second needs to be greater than the value in **(ii)** in order to produce this amount of electrical power.

.....

.....

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..... [2]

3

**(a)** State Hooke's Law.

.....

.....[1]

**(b)** A spring is compressed by applying a force. Fig. 6.1 shows the variation of the force $F$ with compression $x$.

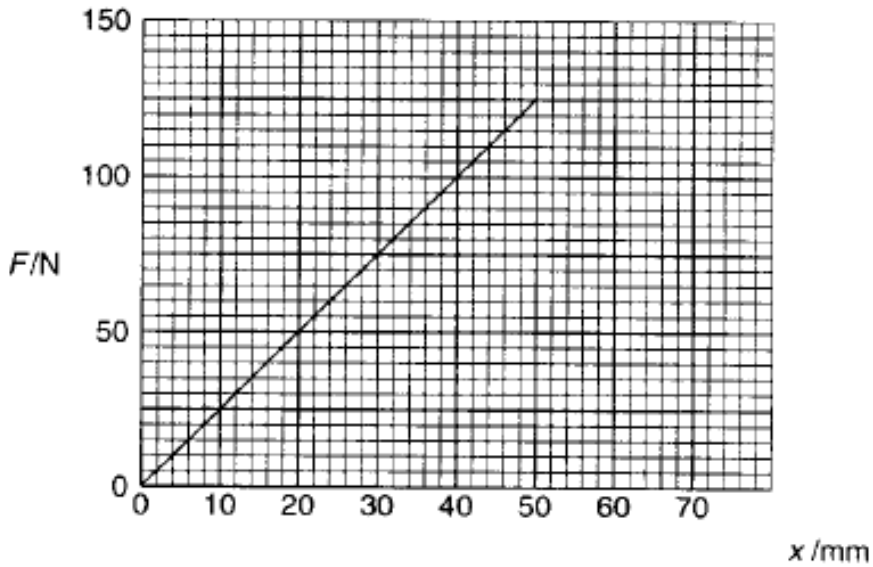


**Fig. 6.1**

**(i)** Calculate the spring constant.

spring constant = ..... unit ..... [2]

**(ii)** Show that the work done in compressing the spring by 48 mm is 2.9 J.

[2]

**(c)** Fig. 6.2 shows the spring in a toy gun. The spring is used to fire a dart of mass 15 g vertically.

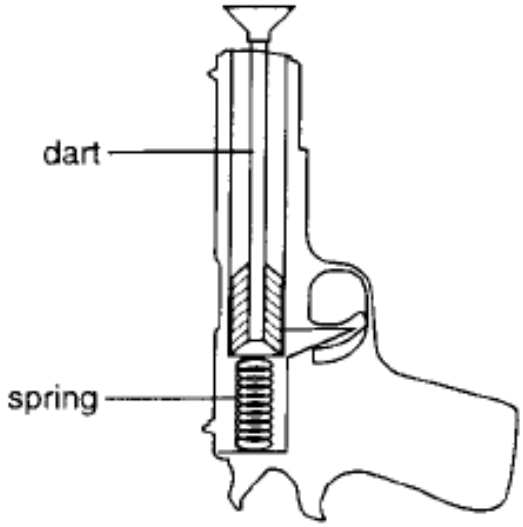


**Fig. 6.2**

**(i)** The spring is compressed by 48 mm in the gun. When the gun is fired the strain energy in the spring is converted into the kinetic energy of the dart. Calculate the speed with which the dart initially leaves the spring when the gun is fired.

speed = ..... $m\,s^{-1}$ [3]

**(ii)** Give two reasons why the dart is unlikely to have 2.9 J of gravitational potential energy when it reaches its maximum height.

1. .....

.....

2. .....

.....[2]

This question is about pressing a red hot bar of steel into a sheet in a rolling mill.

(a) A bar of steel of mass 500 kg is moved on a conveyor belt at 0.60 m s$^{-1}$.
Calculate the momentum of the bar giving a suitable unit for your answer.

momentum = ..... unit ..... [2]

(b) From the conveyor belt, the bar is passed between two rollers, shown in Fig. 1.1. The bar enters the rollers at 0.60 m s$^{-1}$. The rollers flatten the bar into a sheet with the result that the sheet leaves the rollers at 1.8 m s$^{-1}$.

(i) Explain why there is a resultant horizontal force on the bar at the point immediately between the rollers.

.....

.....

.....

.....[2]

(ii) The original length of the bar is 3.0 m. Calculate the time it takes for the bar to pass between the rollers.

time = ..... s [1]

**iii** Calculate the magnitude of the resultant force on the bar during the pressing process.

force = ..... N [3]

5 This question is about the interactions between three identical perfectly elastic solid cubes.

**(a)** Fig. 1.1 shows three numbered cubes each of mass $m$ placed in contact on a frictionless horizontal surface. A steady horizontal force $P$ is applied to the end surface of cube **1**.

**Fig. 1.1**

**(i)** Show that the acceleration of cube 3 is $\frac{P}{3m}$.

[1]

**(ii)** Write down an expression for the resultant force $F_3$ on cube **3**.

$F_3$ = .....[1]

**(iii)** Write down expressions in terms of $P$ for

the acceleration $a_2$ of cube **2**

$a_2$ = .....

the resultant force $F_2$ on cube **2**.

$F_2$ = .....[2]

**(iv)** Hence write down an expression for the magnitude of the force applied by

cube **3** on cube **2**, $F_{32}$

$F_{32}$ = .....

cube **1** on cube **2**, $F_{12}$.

$F_{12}$ = .....[2]

**(b)** Cube **3** is removed. Cube **1** is moved to the left and then projected towards the stationary cube **2** with speed $u$. Suppose that after collision cube **1** moves with speed $v_1$ and cube **2** moves with speed $v_2$. See Fig. 1.2.

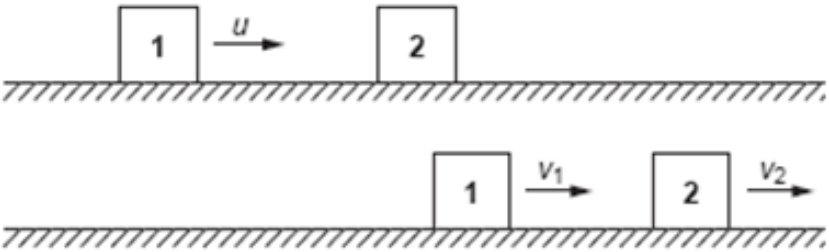


**Fig. 1.2**

(i) Write down equations for the conservation of momentum and of energy in the elastic collision in terms of $m$, $u$, $v_1$ and $v_2$.

equation 1 for momentum .....

equation 2 for energy .....[2]

(ii) Put the values $v_1 = 0$ and $v_2 = u$ into your equations in **(i)** to show that they are solutions.

[1]

**(c)** Cube **3** is replaced in its original position close to cube **2**. Cube **1** is moved to the left and projected towards cube **2** with speed $u$. See Fig. 1.3.

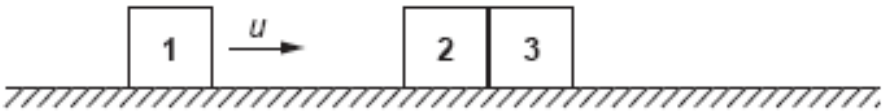


**Fig. 1.3**

(i) After the collision, cubes **1** and **2** remain at rest and cube **3** moves with speed $u$. Explain these observations.

.....

.....

.....

.....[2]

(ii) Cubes **2** and **3** are now glued together. Describe without calculation or explanation what happens in this situation when cube **1** is projected towards cube **2** as in Fig. 1.3.

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.....

.....[2]

A boy on a board B slides down a slope, as shown in Fig. 3.1.

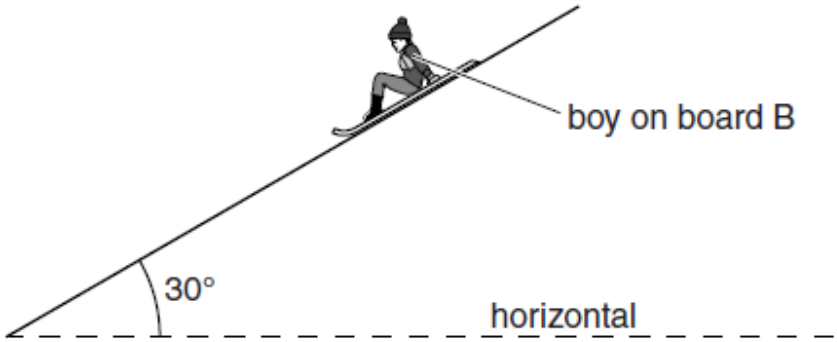


**Fig. 3.1**

The angle of the slope to the horizontal is 30°. The total resistive force *F* acting on B is constant.

**(i)** State a word equation that links the work done by the force *F* on B to the changes in potential and kinetic energy.

.....

.....[1]

**(ii)** The boy on the board B moves with velocity $v$ down the slope. The variation with time $t$ of $v$ is shown in Fig. 3.2.

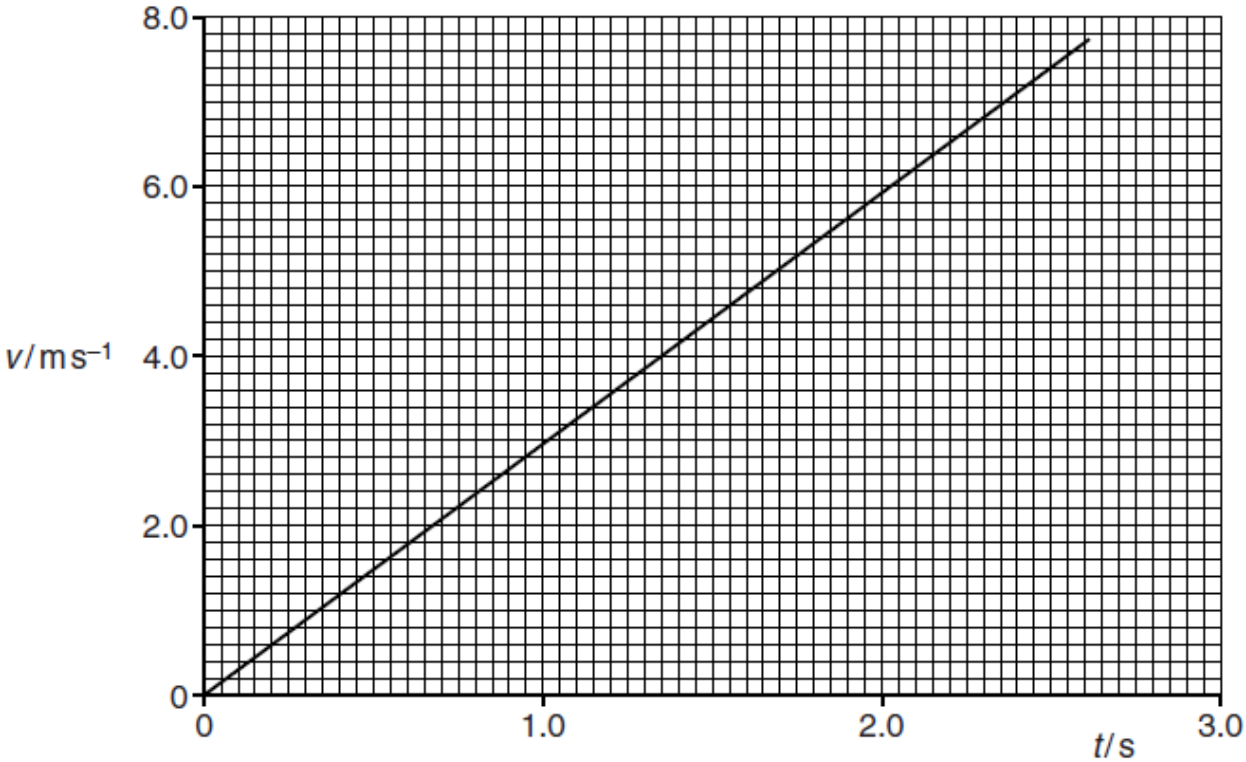


**Fig. 3.2**

The total mass of B is 75 kg.
For B, from $t = 0$ to $t = 2.5$ s,

**1.** show that the distance moved down the slope is 9.3 m,

[2]

**2.** calculate the gain in kinetic energy,

gain in kinetic energy = ..... J [3]

**3.** calculate the loss in potential energy,

loss in potential energy = ..... J [3]

**4.** calculate the resistive force *F*.

*F* = ..... N [3]

7

A spring hangs vertically from a point P, as shown in Fig. 4.1.

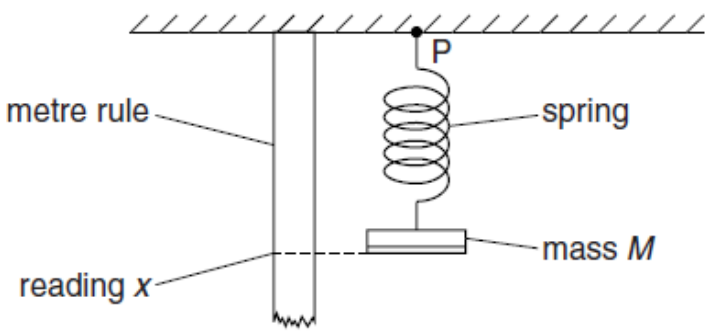


**Fig. 4.1**

A mass $M$ is attached to the lower end of the spring. The reading $x$ from the metre rule is taken, as shown in Fig. 4.1. Fig. 4.2 shows the relationship between $x$ and $M$.

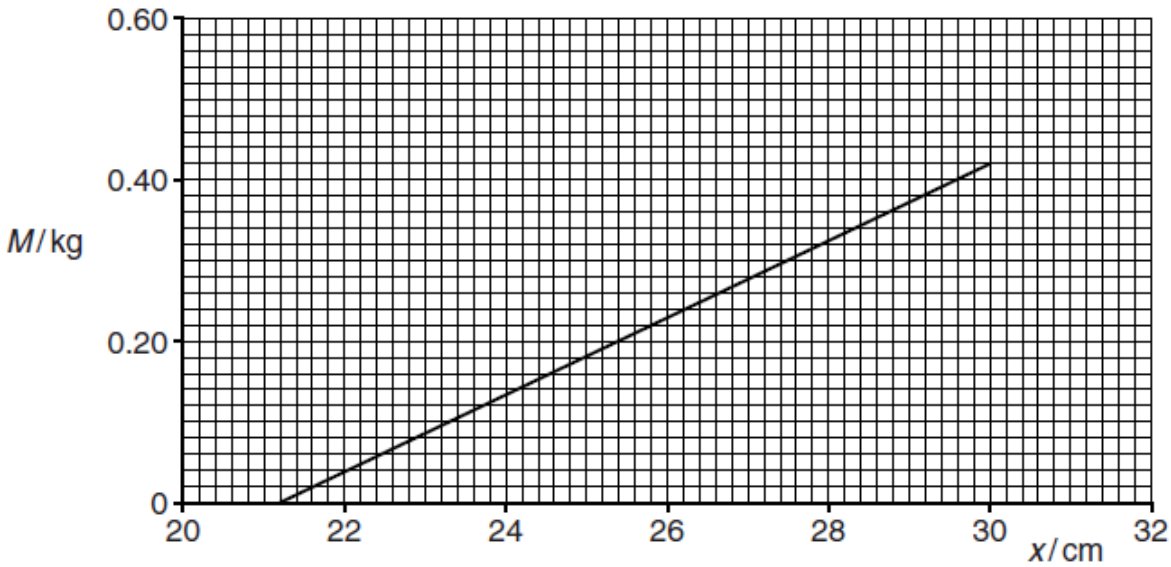


**Fig. 4.2**

**(a)** Explain how the apparatus in Fig. 4.1 may be used to determine the load on the spring at the elastic limit.

.....[2]

**(b)** State and explain whether Fig. 4.2 suggests that the spring obeys Hooke's law.

.....[2]

**(c)** Use Fig. 4.2 to determine the spring constant, in $N\,m^{-1}$, of the spring.

spring constant = ..... $N\,m^{-1}$ [3]

8

Explain what is meant by *gravitational potential* energy and *kinetic* energy.

gravitational potential energy: .....

.....

kinetic energy: .....

.....

[2]

9

The Young modulus of the metal of a wire is 0.17 TPa. The cross-sectional area of the wire is 0.18 $mm^2$.

The wire is extended by a force *F*. This causes the length of the wire to be increased by 0.095 %.

Calculate

**(i)** the stress,

stress = ..... Pa [4]

**(ii)** the force *F*.

*F* = ..... N [2]

A motor drags a log of mass 452 kg up a slope by means of a cable, as shown in Fig. 2.1.

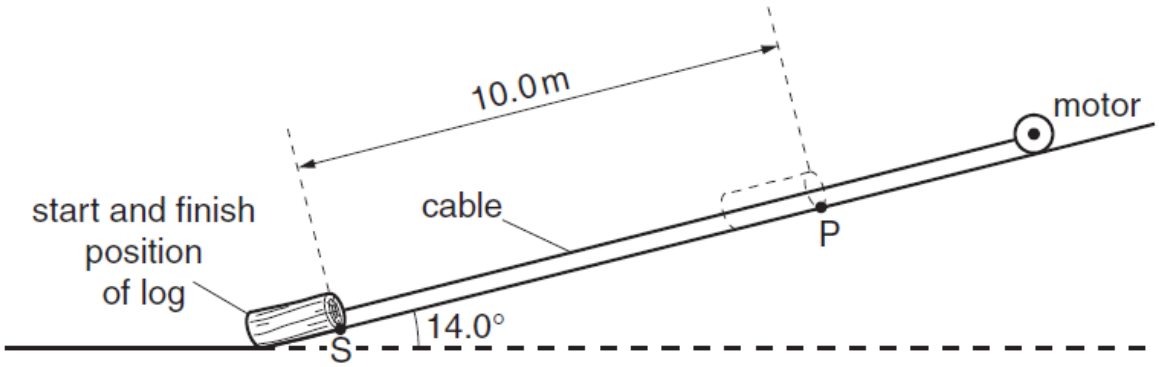


**Fig. 2.1**

The slope is inclined at 14.0° to the horizontal.

**(a)** Show that the component of the weight of the log acting down the slope is 1070 N.

[1]

**(b)** The log starts from rest. A constant frictional force of 525 N acts on the log. The log accelerates up the slope at $0.130\,\mathrm{m\,s^{-2}}$.

**(i)** Calculate the tension in the cable.

tension = ..... N [3]

**(ii)** The log is initially at rest at point S. It is pulled through a distance of 10.0 m to point P.

Calculate, for the log,

**1.** the time taken to move from S to P,

time = ..... s [2]

**2.** the magnitude of the velocity at P.

velocity = ..... $m\,s^{-1}$ [1]

11

A weight of 7.0 N hangs vertically by two strings AB and AC, as shown in Fig. 2.1.

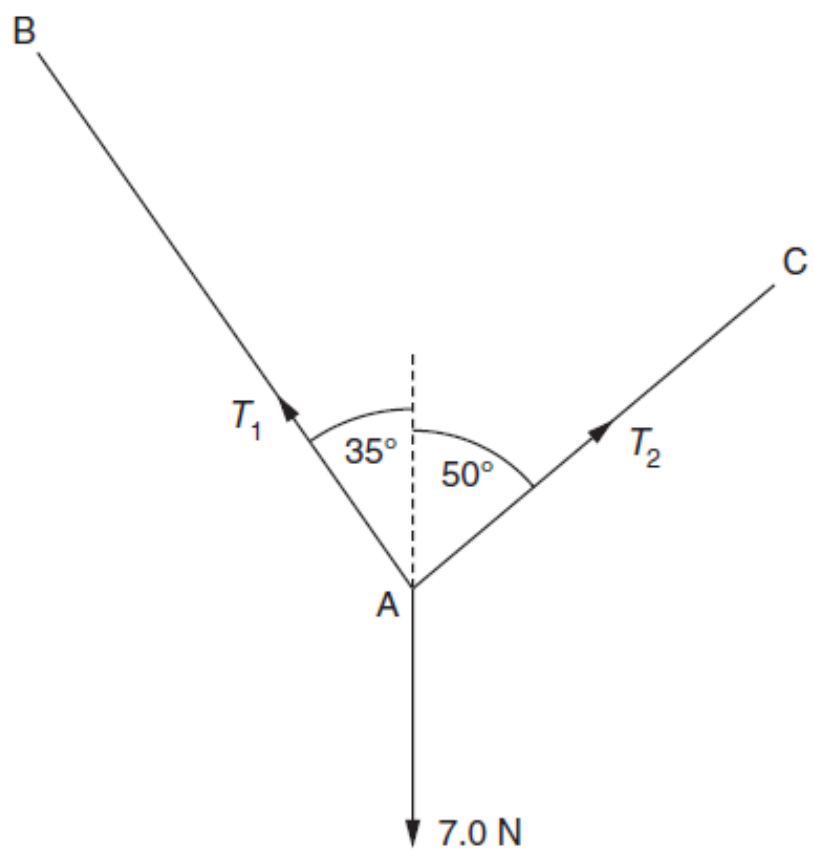


**Fig. 2.1**

For the weight to be in equilibrium, the tension in string AB is $T_1$ and in string AC it is $T_2$.

On Fig. 2.1, draw a vector triangle to determine the magnitudes of $T_1$ and $T_2$.

$T_1$ = ..... N

$T_2$ = ..... N

[3]